



A review on synergizing knowledge graphs and large language models

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Abstract

This paper examines the integration of large language models (LLMs) with knowledge graphs (KGs) through a systematic four-layer framework that includes data, model, technology, and application dimensions. We analyze the capabilities and limitations of LLMs in natural language processing along with the strengths and challenges of KGs in knowledge representation. We address fundamental weaknesses in each approach and identify complementary integration methods. Our analysis reveals that LLMs excel at contextual understanding and generation but struggle with factual consistency and reasoning transparency. In contrast, KGs provide structured and verifiable knowledge but lack adaptability to unstructured inputs. We review integration strategies, including knowledge injection techniques, retrieval-augmented generation, and neuro-symbolic approaches. The combined methods demonstrate significant performance improvements. Through the case study of the GLM architecture, we demonstrate how the integration of KGs and LLMs improves accuracy, interpretability, and factual grounding in specialized domains and also shows substantial performance improvements in knowledge-intensive tasks (15–20% on MedQA and 14–17% on the medical MMLU benchmarks). The resulting hybrid systems offer concrete advantages in critical applications requiring precision and adaptability, including healthcare diagnostics, financial compliance, and educational technology. Lightweight knowledge representation, adaptive update mechanisms, and unified cross-modal frameworks are promising research directions to advance KG–LLM integration.

Keywords Large language models (LLMs) · Knowledge graphs (KGs) · Knowledge generation · GLM

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1 Introduction

Large language models (LLMs) have revolutionized Natural Language Processing (NLP) by demonstrating exceptional capabilities in understanding and generating human-like text. Models such as GPT-3 [1] and ChatGPT¹ showcase cross-domain knowledge integration, adaptability, and versatility in applications ranging from text summarization to complex question answering [2]. However, these models face critical challenges. Their reasoning processes are often opaque [3], rendering them “black-box” systems with limited interpretability [4]. Moreover, their tendency to hallucinate (generating plausible but factually incorrect information) poses significant risks [5], especially in high-stakes domains such as healthcare or legal advice. Updating knowledge to remain aligned with evolving data also remains an open challenge [6].

Consider a healthcare scenario where an LLM is asked about drug interactions with warfarin. Standard LLMs might generate a comprehensive list that could omit critical interactions or include incorrect ones based solely on statistical correlations in training data. Such hallucinations could have serious clinical consequences. Furthermore, the model cannot cite specific medical references or explain its reasoning process, creating a trust barrier that limits verification.

In contrast, Knowledge Graphs (KGs) provide structured representations of knowledge through entities and relationships, enabling machines to perform precise reasoning and high-confidence queries. By nature, KGs offer interpretability, transparency, and factual accuracy, making them well suited for applications requiring trustworthiness and domain-specific expertise. However, KGs struggle with understanding and processing unstructured natural language and face challenges in maintaining relevance as knowledge evolves.

The complementary strengths of LLMs and KGs make their integration a promising approach [7]. Although LLMs excel in unstructured data processing with rich linguistic context and adaptability, KGs contribute structured, interpretable knowledge that grounds reasoning in factual accuracy. Together, these technologies can address each other's limitations, paving the way for AI systems that are more capable, reliable, explainable, and adaptable to real-world complexities.

Previous reviews have addressed individual aspects of LLM or KG. Ji et al. conducted a comprehensive review of knowledge graph representation, acquisition, and application, laying the foundation for subsequent research [8]. Yin et al. focused on knowledge-intensive NLP and pre-trained language models, emphasizing effective knowledge shaping to improve task-specific performance [9]. Jiang et al. systematically examined the development of knowledge graphs, providing unique perspectives on the adaptation of KG in the era of large language [10]. More recently, Pan et al. proposed a roadmap for unifying large language models with knowledge graphs [11], while Fichtl et al. detailed technical implementations of knowledge-enhanced language models [12]. Ibrahim et al. comprehensively discussed the

¹ <https://openai.com/index/chatgpt/>.

challenges and opportunities of using LLMs to improve KGs from the perspective of model construction, evaluation, and benchmarking [7].

This paper uniquely provides a comprehensive layer-by-layer analysis from fundamental data structures to practical applications. By organizing our review according to a four-layer framework (Data, Model, Technology, and Application layers), we offer researchers and practitioners a structured understanding of LLM-KG integration's current state, challenges, and potential. The paper makes several key contributions: (1) systematically analyzing LLM-KG integration through a novel four-layer framework (data, model, technology and application layers) and providing a comprehensive and structured understanding of their synergistic relationship; (2) identifying and comparing various integration approaches across all layers while providing customized solutions to technical challenges encountered at every level; (3) presenting a detailed case study of GLM that demonstrates how the four-layer framework manifests in practice, with quantitative evidence of significant performance improvements in knowledge-intensive tasks.

The paper is structured as follows. Section 2 contrasts the capabilities and limitations of LLMs and KGs, highlighting their complementary nature. Section 3 details the integration approaches across the four-layer framework: the Data Layer examines how LLMs enhance KG construction and how KGs improve LLM training; the Model Layer explores architectural approaches to integrate KG knowledge within LLMs; the Technology Layer reviews technical integration mechanisms; and the Application Layer demonstrates improved performance in domains such as healthcare, education, and finance. Section 4 presents our GLM case study. Section 5 presents the conclusion.

2 KGs versus LLMs

2.1 Brief introduction of KGs

A knowledge graph is a structured graph to represent knowledge, designed to capture entities and their interrelationships in a graph. The basic building blocks of a knowledge graph include entities (nodes) and relationships (edges), as well as attribute and value pairs [13]. In a knowledge graph, entities act as nodes that represent concrete or abstract concepts in the real world, such as people, places, events, or ideas. These entities are connected through relationships, forming the graph's edges and describing various connections and interactions between entities. Each entity can also have properties that store additional descriptive information in key-value pairs. These three elements - entities, relations and attributes - combine to form the basic unit of the knowledge graph [14]: the triplet $\mathcal{KG} = \{(h, r, t) \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}\}$. Triples are usually expressed as (subject, predicate, object) forms, e.g. (Shakespeare, Creation, Hamlet).

As shown in Table 1, the traditional rule-based method relies on manual or semi-automated knowledge extraction processes, so its relational reasoning ability is relatively limited and its update mechanism is relatively static, making it difficult to adapt to new changes. However, it is very interpretable because of its clear definition

Table 1 Dynamic construction methods of KGs

Characteristics	Methods				
	Traditional	Statistics	Early	Trans-	LLMs
	Rule		NN	former	Dynamic
Knowledge extraction	Manual	Partial	Elementary	Advanced	Full
Relation inference	Limited	Partial	Simple	Complex	Deep
Renewal mechanism	Static	Periodic	Discrete	Incremental	Real-time
Domain adaptability	Low	Medium	Medium	High	Very High
Computational efficiency	High	Medium	Low	Medium	Low
Interpretability	Excellent	Common	Poor	Average	Well

of rules. Although the statistics-based method has achieved automated knowledge extraction to some extent, its relational reasoning ability is still not comprehensive enough. The methods require regular updates to keep the data up-to-date. Its computational efficiency is moderate, but its interpretability is relatively ordinary due to the lack of an intuitive interpretation framework.

Early neural network methods have initially achieved automated knowledge extraction, but their relational reasoning ability is relatively simple. The update mechanism, which requires discrete adjustments, is not flexible enough. Although it has good adaptability in some fields, the complexity of the model causes low computational efficiency and poor interpretability. Transformer-based methods represent the advanced level of current knowledge extraction technology, enabling highly automated knowledge extraction and complex relational reasoning. Supporting an incremental update mechanism can absorb new knowledge without retraining the entire model. However, this method has a high computational cost, but its interpretability is still better than that of early neural network methods.

Currently, one of the most advanced automated knowledge extraction methods is the LLMs dynamic construction method. It can not only fully automatically extract knowledge, but also perform in-depth relational reasoning. More importantly, it supports a real-time update mechanism that enables the model to respond quickly to the latest information and changes. Although this method has certain challenges in terms of computational efficiency, its high domain adaptability and good interpretability make it the preferred solution for many complex tasks.

2.2 Brief introduction of LLMs

LLMs represent a significant advance in NLP, characterized by their huge parameter sizes that range from billions to hundreds of billions. These sophisticated deep learning models are designed to understand and generate human-language text across diverse tasks including translation, summarization, question-answering, and text generation.

Figure 1 illustrates key milestones in LLM development from 2018 to 2024. In 2018, Google introduced BERT [15], which features a bidirectional Transformer

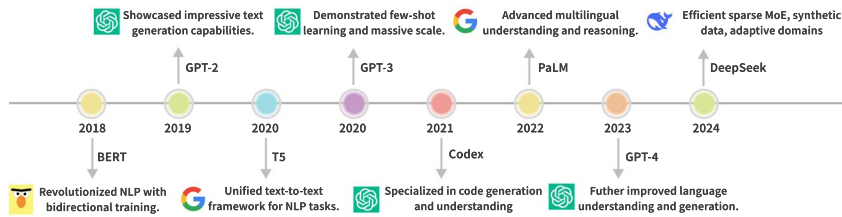


Fig. 1 Breakthroughs in LLM development (2018–2024)

architecture with 110–340 million parameters that revolutionized masked language modeling and improved semantic understanding. The following year, OpenAI’s GPT-2 (1.5 billion parameters) demonstrated impressive text generation capabilities through its transformer decoder structure [16].

The year 2020 marked a pivotal moment with two significant developments. OpenAI’s GPT-3 [1] established a new scale benchmark with 175 billion parameters, enabling few-shot learning capabilities across diverse tasks, while Google’s T5 [17] unified NLP tasks through a text-to-text framework with parameters ranging from 11 to over 100 billion.

In 2021, OpenAI’s Codex [18] specialized in code generation and understanding, driving tools like GitHub Copilot through optimized training on programming data. Google’s PaLM [19] followed in 2022 with 540 billion parameters and a Pathways architecture that enabled exceptional performance in logical reasoning and multilingual applications. In 2023, OpenAI’s GPT-4 [20] advanced language understanding and generation through architectural refinements and multimodal integration.

DeepSeek emerged in 2024 as an innovative AI system featuring a Hybrid Expert Architecture (MoE) that dynamically allocates computing resources across specialized sub-models. This approach, combined with its Multi-Head Latent Hotspot (MLA) technology for efficient long-text processing, significantly increased parameter capacity while reducing computational costs. DeepSeek [21] also pioneered a minimalist reinforcement learning paradigm through its R1 model, stimulating autonomous reasoning capabilities and providing new directions for general artificial intelligence development.

Throughout this evolution, LLMs have undergone remarkable transformation: parameter scales have expanded from hundreds of millions to trillions, architectures have evolved from unidirectional decoding and bidirectional encoding to sophisticated multimodal integration, and application domains have broadened from text understanding to code generation, mathematical reasoning, and cross-modal interactions - continuously pushing the boundaries of artificial intelligence capabilities.

2.3 KGs versus LLMs

There are fundamental differences in the reasoning mechanisms of knowledge graphs and large language models, which pose both integration challenges and complementary opportunities. KGs adopt a reasoning method based on symbolic logic,

supporting deterministic reasoning paths and formal verification through an explicit rule system [22]. This approach allows systems to perform precise and interpretable reasoning with highly verifiable results. KGs can verify correctness through formal proofs, making them particularly suitable for scenarios requiring strict logical guarantees.

In contrast, LLMs use probabilistic reasoning based on statistical patterns in training data, achieving implicit knowledge representation through parameter weights [23]. When generating answers, LLMs rely on learned probability distributions, a process closer to pattern recognition than logical deduction. Although this enables handling of fuzzy and uncertain situations, it creates “black-box” characteristics that make it difficult to provide clear reasoning foundations.

This fundamental difference creates both challenges and opportunities for integration. The challenge lies in coordinating rule-based deterministic reasoning with statistical-based probabilistic approaches: how to introduce the KG rigor while maintaining LLM flexibility [24]. These challenges have spawned innovative architectures like Neuro-Symbolic Systems that attempt to combine advantages from both paradigms.

The complementary opportunities are significant: LLMs excel at common sense reasoning and handling fuzzy situations difficult to formalize in KGs, while KGs provide accurate factual bases and verifiable reasoning chains. Through integration, systems can prioritize the reasoning of KG for precise answers (in medical diagnosis or legal reasoning) and leverage the flexibility of LLM for fuzzy problems (content creation or sentiment analysis), achieving a deep integration of “symbolism + connectionism”.

As shown in Table 2, KGs and LLMs have distinct functional characteristics and application scenarios [25]. KGs offer strong interpretability with clear logic, but face challenges in knowledge acquisition and updates. They excel in domain expertise, but struggle with natural language understanding. LLMs provide multi-task generalization and strong generative capacity, but may produce hallucinations and lack interpretability [26].

This integration trend is becoming increasingly significant. Knowledge-enhanced language models can reduce hallucinations by incorporating structured relationships, while LLM-driven knowledge engineering can efficiently extract entities and relationships from text. Future integration architectures that combine symbolism and connectionism will balance efficiency, explainability, and flexibility, opening new paths for artificial intelligence development.

3 Enhanced LLMs and knowledge graphs integration

This review considers three primary types of KGs: (1) general purpose (e.g. Wikidata), (2) domain specific (e.g., UMLS for medicine), and (3) enterprise knowledge graphs. For LLMs, we examine decoder-only models (GPT family), encoder-decoder models (T5), and encoder-only models (BERT) with scale variations from 1B to 175B parameters.

Table 2 Comparison of KGs and LLMs

Dimensions	KGs	LLMs
Interpretability	Structured knowledge, clear logic, traceable, strong interpretability	Black box model, hard to trace, weak interpretability
Universality	Deep domain knowledge but limited universality	Multi-task generalization, cross-language and multi-modal support
Generative capacity	Knowledge query and reasoning, cannot generate natural content	Generates text, code, images with creative flexibility
Knowledge updating	Manual maintenance, costly and time-consuming updates	Supports incremental training/fine-tuning (risk of catastrophic forgetting)
Common sense	Relies on structured knowledge, lacks unencoded common sense	Strong contextual reasoning from corpus learning
Credibility	Expert-reviewed reliability, traceable/correctable errors	May contain hallucinations, errors hard to trace
Field expertise	High-precision in vertical fields, complex reasoning	Good in general scenarios, needs external knowledge for specialized fields
Resources	Requires domain experts, costly construction	Pre-trained, minimal fine-tuning, high resource reuse

The bidirectional integration of knowledge graphs (KGs) and large language models (LLMs) can be effectively achieved through a collaborative framework that spans the data, model, technology and application layers [11]. This integration is crucial to improving the accuracy and interpretability of AI models. Figure 2 provides a comprehensive visualization of the multi-layered framework for integrating Knowledge Graphs with Large Language Models, illustrating how knowledge flows from the foundational Data Layer through the Model and Technical Layers to reach practical applications, with the mid-integration approach serving as the connecting mechanism across all components of the framework.

3.1 Data layer in KELMs

The design and implementation of knowledge-enhanced large models (KELM) [12] at the data layer is the foundation. This level involves the collection, modeling, storage, processing, and integration of data in larger models, with the goal of enabling models to efficiently use external knowledge and improve the accuracy, relevance, and interpretability of the generated results. The following are detailed in several aspects.

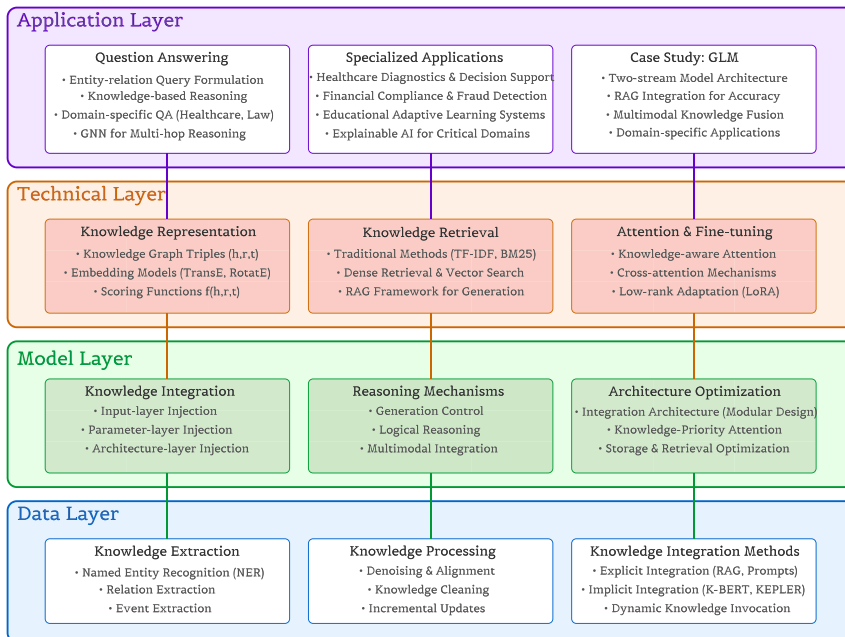


Fig. 2 Enhanced LLMs and knowledge graphs integration architecture (highlighted areas represent mid-integration layers)

3.1.1 Data processing

The data processing layer is critical to ensure that the knowledge integrated in KELM is accurate, relevant, and up-to-date. Knowledge extraction serves as the initial step, involving the identification of meaningful entities, relationships, and events from raw data. Techniques such as named entity recognition (NER) [27] identify entities (e.g. persons, organizations, locations), while relation extraction determines the relationships between these entities (e.g. “works at”, “is located in”), structuring knowledge into usable forms. Event extraction [28], designed to handle dynamic and complex knowledge, captures information about specific events (e.g. company mergers, medical procedures, geopolitical events), often relying on sophisticated algorithms.

Subsequent steps involve knowledge cleaning, which addresses errors, redundancies, and irrelevant data. Denoising prevents noisy or contradictory information from contaminating the knowledge base [29], while alignment techniques (e.g. string matching, embedding-based similarity) consolidate duplicate entities across sources to ensure consistency [30]. Finally, updating knowledge maintains the relevance of the integrated knowledge. For dynamic domains (e.g., financial markets, news), real-time updates are crucial, while incremental updates incorporate new data periodically to improve the completeness of the knowledge base [31]. This robust pipeline ensures that the knowledge base remains accurate, rich, and adaptable to evolving contexts.

3.1.2 Integration of knowledge and large language models

Integrating knowledge into KELMs can be achieved through explicit and implicit methods, as well as dynamic knowledge invocation.

Explicit integration uses external knowledge sources directly to improve model performance [32]. One popular technique, RAG [33], involves retrieving relevant knowledge (e.g., from knowledge graphs or text corpora) and using it to generate responses, thus improving the accuracy and relevance of outputs. Another explicit integration method is prompt enhancement, where contextual information or entity descriptions are directly included in input prompts, providing the model with the necessary knowledge upfront [34].

Implicit integration integrates knowledge into the model’s internal representations during training. For example, knowledge-enhanced models like K-BERT [35] and KEPLER [36] inject knowledge embeddings into the model during pretraining, allowing it to internalize explicit knowledge from external sources. Fine-tuning domain-specific tasks with knowledge-augmented data further adapts the model to specific requirements while leveraging this integrated knowledge for improved performance [37].

Dynamic Knowledge Invocation allows the model to access and utilize knowledge adaptively during inference [38]. This approach includes dynamic prompting, where the model constructs context-aware prompts in real time, and knowledge reasoning modules [39], which verify or supplement the model’s internal knowledge

with external sources during response generation, enhancing accuracy and contextual relevance.

These strategies enable KELMs to effectively combine explicit and implicit knowledge integration with dynamic adaptability, addressing tasks that require domain expertise and nuanced contextual understanding.

3.1.3 Challenges and optimization directions

Based on the research paper, the data layer of KELMs faces several interconnected challenges. Data quality issues including noise, redundancy, and contradictory information directly impact model reliability, requiring knowledge cleaning techniques like denoising and entity alignment combined with dynamic verification mechanisms. Cross-domain knowledge integration is complicated by structural and semantic differences between knowledge graphs in different domains, necessitating unified representation frameworks such as cross-modal graph embedding to maintain domain specificity while achieving semantic interoperability. As knowledge bases grow, scale and efficiency bottlenecks emerge where traditional storage and retrieval methods become inadequate, calling for solutions that combine vector databases with graph indexing technology and incremental update mechanisms. In time-sensitive domains, insufficient knowledge dynamics lead to outdated information, requiring real-time data pipelines and dynamic knowledge retrieval to enable quick responses to new information. These challenges are a “quality-scale-timeliness triangle contradiction” that must be balanced, our future research will explore lightweight knowledge representation and adaptive update strategies to enable models to maintain massive knowledge while dynamically refreshing outdated information.

3.2 Model layer in KELMs

In the design of KELMs, the model layer is the core that determines how knowledge and large language models (LLMs) are integrated to improve task performance. The model layer mainly involves the knowledge injection mode, reasoning mechanism, generation process, and optimization of the model architecture design. The specific contents are as follows: interpretability, and generalization capabilities.

3.2.1 Knowledge integration and reasoning mechanisms

KELMs achieve dynamic utilization of external knowledge through coordinated knowledge injection and reasoning generation mechanisms. Knowledge injection occurs at three levels: input-layer (through prompt engineering or RAG), parameter-layer (like K-BERT [35] aligning knowledge triples via attention mechanisms), and architecture-layer (such as K-Adapter’s [40] independent knowledge module). These approaches provide multilevel knowledge support from real-time data to internalized common sense.

The reasoning generation mechanism focuses on knowledge application during decoding. Generation control ensures logical consistency through knowledge

constraints and dynamic prompt updates [41]. Logical reasoning combines explicit rules with implicit parameterized knowledge for multilevel inference [42]. Multimodal integration improves context perception through cross-modal knowledge graphs [43]. These techniques collaborate in architectural design - RAG invokes parameterized knowledge while retrieving external knowledge, while K-Adapter supports incremental learning through dedicated knowledge processing.

A core technical challenge is effectively aligning structured knowledge graph representation with the semantic space of language models. Key alignment methods include:

Knowledge-Aware Attention: Models like KAFT enable dynamic focus on relevant knowledge during generation through entity-aware attention weights. The approach modifies standard Transformer attention calculation with an entity relationship matrix, like Eq. (1):

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + E_{\text{rel}}\right)V, \quad (1)$$

Where E_{rel} represents the entity relationship matrix extracted from the knowledge graph, which directly affects the attention distribution, allowing the model to prioritize the associated information in the knowledge graph when processing content related to a specific entity. This mechanism is particularly suitable for entity-intensive tasks such as named entity recognition and relation extraction.

Cross-Modality Embedding Spaces is another powerful alignment method, with KEPLER as a typical representative. KEPLER optimizes language modeling and knowledge embedding objectives simultaneously to create a shared semantic space. Its training objective function can be expressed as Eq. (2):

$$L = \lambda_{\text{MLM}} \cdot L_{\text{MLM}} + \lambda_{\text{KE}} \cdot L_{\text{KE}} \quad (2)$$

Where L_{MLM} is the masked language modeling loss, L_{KE} is the knowledge embedding loss (such as TransE loss), and λ_{MLM} and λ_{KE} are hyperparameters that weigh the two objectives. Through this joint training, entity representation and word representation are mapped to the same vector space, allowing the model to seamlessly switch between text understanding and knowledge reasoning. The advantage of this method is that it deeply integrates the structural information of the knowledge graph and the semantic understanding ability of the language model.

Retrieval-Augmented Approaches such as RAG adopt a more flexible alignment strategy by coordinating the vector representations between the knowledge graph and the hidden state of the language model. The RAG model consists of two key components: the retriever and the generator. The retriever maps the query to a vector representation to retrieve relevant information from the knowledge base; the generator generates answers based on the retrieved information and the original query. The key alignment occurs in the retrieval phase, where the similarity calculation between the query vector q and the knowledge graph entity vector e can be expressed as Eq. (3):

$$\text{sim}(q, e) = q^T \cdot e / (\|q\| \cdot \|e\|) \quad (3)$$

To optimize this similarity calculation, RAG usually uses contrastive learning to make the query vector generated by the language model close to the relevant knowledge entity vector and away from the irrelevant entity vector. The advantage of this method is that it maintains the generation flexibility of LLM while being able to introduce the structured knowledge of KG on demand.

In addition, emerging neurosymbolic approaches, such as KagNet, achieve finer-grained alignment by explicitly modeling the relationship between the reasoning path of the knowledge graph and the attention mechanism of the language model. KagNet first extracts possible reasoning paths from the knowledge graph, then encodes these paths using a graph neural network, and finally fuses the path representation with the hidden state of the language model through the attention mechanism. This method is particularly suitable for complex problems that require multi-hop reasoning.

These alignment methods have distinct advantages for different scenarios: knowledge-aware attention for entity-intensive tasks, cross-modal embedding for general language understanding, retrieval-augmented methods for open-domain QA requiring external knowledge, and neurosymbolic methods for professional applications needing precise reasoning. Practical implementation often combines multiple methods based on specific task requirements.

3.2.2 Model architecture optimization

In KELMs, optimizing the architecture is crucial to efficiently integrate and utilize external knowledge. This requires designing the model in a way that enhances knowledge integration, prioritizes knowledge retrieval, and optimizes storage and retrieval systems.

Integration architecture Modular design and multitask joint learning work together as complementary approaches in KELMs [44]. By implementing a modular architecture that separates the components of knowledge processing from the core language model, the system gains flexibility to update and extend its knowledge base without affecting fundamental operations [45], while simultaneously employing multitask joint learning to train the model in both knowledge identification and language processing tasks [46]. This integrated approach ensures that knowledge can be continuously updated and expanded with domain-specific data [47], while parallel training improves the model's ability to extract entities and relationships from text, ultimately leading to better knowledge integration and application across diverse contexts [48].

Knowledge-priority attention mechanisms Relation-aware attention [49] and hierarchical attention [50] mechanisms serve as sophisticated approaches to knowledge processing in language models. These complementary mechanisms work together by allowing models to focus on key relationships within knowledge graphs while simultaneously assigning different weights based on information importance [51]. By allowing the model to prioritize crucial connections (such as “symptom of” or “treatment for” in medical contexts) and efficiently filter out less relevant

information, these attention mechanisms help ensure both precision in relationship handling and optimal knowledge utilization during the generation or reasoning process. This dual attention approach creates a more nuanced and effective way to process and apply knowledge, leading to more accurate and contextually appropriate outputs.

Storage and retrieval optimization Parameter-efficient methods and external storage interaction mechanisms work in tandem to optimize knowledge management in language models [52]. Using techniques such as LoRA [53] and Adapter [54] layers for dynamic knowledge loading, while simultaneously leveraging external storage systems such as vector and graph databases for real-time knowledge retrieval, these models achieve both computational efficiency and knowledge accessibility. This combined approach allows efficient integration of knowledge during inference without full model retraining, while ensuring access to current information through external storage interaction [55]. Together, these architectural optimizations enable models to maintain high performance with continuously updated knowledge, ultimately producing more accurate and contextually relevant outputs in a resource-efficient manner.

3.2.3 Challenges and optimization directions

The model layer faces several key challenges: ambiguity and conflicts of knowledge, insufficient coverage of vertical domain knowledge, and low real-time reasoning efficiency. To address these issues, knowledge verification modules should be introduced for multi-source cross-validation, combined with rule priority mechanisms to dynamically screen high-confidence knowledge. Domain-adaptive architectures should be designed to support an incremental injection of expert-verified vertical knowledge. Through knowledge representation compression, precalculation of high-frequency relationship paths, and optimization of GPU-CPU collaborative computing architecture, pipeline parallel processing of “searching and generating simultaneously” can be achieved. Combining symbolic logic with attention interpretability analysis builds a dual-channel interpretation system, enhancing model robustness against noise and adversarial samples. The future evolutionary direction of the model layer is the “symbiosis of symbols and connections”, which requires the breakthrough of neural symbolic system bottlenecks to achieve models with rigorous reasoning capabilities, such as expert systems and flexible generation, such as LLM, ultimately realizing “white-box intelligence”.

3.3 Technical layer in KELMs

From a technical perspective, KELMs combine natural language processing (NLP) capabilities with structured or unstructured external knowledge, such as knowledge graphs, to improve model performance in specialized tasks [56]. The integration of knowledge into these models involves several key technologies and methodologies, including knowledge representation, knowledge retrieval, attention mechanisms, and optimization techniques [8].

Knowledge representation and retrieval form the dual pillars of knowledge-enhanced language models (KELMs), governing how structured knowledge is encoded, accessed, and utilized. Knowledge graphs (KGs) provide the foundational framework by organizing entities and relationships into triples (h, r, t) , where h (head) and t (tail) are entities connected by relation r . Embedding techniques such as TransE and RotatE map these symbolic structures into continuous vector spaces: TransE enforces $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ through a distance-minimization objective, while RotatE extends this by modeling relations as rotations in complex space ($\mathbf{t} = \mathbf{h} \circ \mathbf{r}$). These embeddings enable KELMs to perform latent reasoning, such as predicting tail entities through scoring functions $f(q, r, t)$.

For retrieval, KELMs leverage both traditional methods (e.g., TF-IDF, BM25) and neural approaches. Dense retrieval systems embed queries and documents into shared vector spaces using models like BERT, computing relevance via cosine similarity $\text{Sim}(\mathbf{q}, \mathbf{d})$. Advanced frameworks like RAG dynamically integrate retrieval and generation, fetching top- k knowledge pieces K and conditioning outputs on $P(y|x) = \sum_{k \in K} P(k|x)P(y|x, k)$. This synergy between structured representation and semantic retrieval ensures KELMs access precise, contextually grounded knowledge while maintaining computational efficiency and scalability.

3.3.1 Attention mechanisms

Attention mechanisms play a pivotal role in KELMs by facilitating the effective integration and utilization of external knowledge during inference and reasoning [57]. These mechanisms are engineered to dynamically focus on the most relevant parts of a knowledge base, such as knowledge graphs, and integrate this information with text input. Knowledge-aware attention enhances the model's ability to emphasize critical relationships between entities. For example, in a knowledge graph represented as triples (h, r, t) , where h is the head entity, r is the relationship, and t is the tail entity, the attention weights α_{ij} can be computed to prioritize relationships that are contextually relevant. These weights are typically derived using a scaled dot product mechanism, like Eq. (4):

$$\alpha_{ij} = \text{softmax}\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}\right), \quad (4)$$

where $\mathbf{q}_i$ and $\mathbf{k}_j$ are the query and key vectors associated with the input text and the entities of the knowledge graph, respectively, and d_k is the dimensionality of the key vectors. The resulting attention scores help the model focus on the most pertinent entity relationships, such as “the cause of” or “being a treatment for” depending on the task.

Multimodal attention mechanisms extend this capability to scenarios in which the model integrates text, images, or other data modalities [58]. For example, given a textual input x and an image v , the attention mechanism computes modality-specific attention weights and combines them to create a unified representation, like Eq. (5):

$$\mathbf{z} = \text{Attention}(\mathbf{x}, \mathbf{K}_t) + \text{Attention}(\mathbf{v}, \mathbf{K}_v), \quad (5)$$

where $\mathbf{K}_t$ and $\mathbf{K}_v$ are the keys for the text and image modalities, respectively. This integration ensures that the model can effectively utilize diverse sources of knowledge.

Finally, cross-attention layers in transformer-based architectures such as T5 or BART are tailored to link input text tokens with external knowledge representations [59]. During generation, each token's representation attends both to the sequence of preceding tokens and the embeddings of related knowledge graph entities. The cross-attention output for a token is computed as Eq. (6):

$$\mathbf{o}_t = \text{softmax}\left(\frac{\mathbf{q}_t \cdot [\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_n]}{\sqrt{d_k}}\right) \cdot [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n], \quad (6)$$

where $[\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_n]$ and $[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n]$ are the key and value vectors of knowledge entities, and $\mathbf{q}_t$ is the query vector for the token. This mechanism ensures seamless integration of external knowledge with language tokens, enabling KELMs to generate contextually enriched and factually grounded responses. By unifying these techniques, attention mechanisms significantly enhance the model's ability to reason over structured and unstructured knowledge, making them indispensable in KELMs.

3.3.2 Fine-tuning and transfer learning

Fine-tuning and transfer learning are essential techniques in KELMs to enable effective specialization in domain-specific tasks while leveraging pre-trained general knowledge [12]. The process begins with pretraining, where a model like K-BERT or KEPLER is trained on a large corpus enriched with structured knowledge, such as triples from a knowledge graph $((h, r, t))$ and textual data. During this stage, the model learns a unified representation of text and knowledge, embedding entities (h, t) and relationships (r) into a shared vector space. For example, using embedding-based scoring functions such as $f(h, r, t) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$, the model optimizes its parameters to align textual and relational representations, internalizing factual and relational knowledge across domains.

Once pretrained, the model undergoes task-specific fine-tuning to adapt to specific applications, such as question answering or medical text classification [60]. During fine-tuning, task-relevant datasets are introduced, often enriched with domain-specific knowledge from tailored knowledge bases. This stage incorporates an additional objective function to align the model predictions with task-specific requirements. For example, in a classification task, the loss of fine-tuning can combine a cross-entropy term $L_{\text{CE}} = -\sum_{i=1}^N y_i \log(\hat{y}_i)$, where y_i and $\hat{y}_i$ are true and predicted labels, with a loss of consistency in knowledge $L_{\text{KG}} = \|\mathbf{h} + \mathbf{r} - \mathbf{t}\|^2$ to ensure that the model's output remains consistent with its embedded knowledge.

This approach allows KELMs to specialize in fields like healthcare or legal reasoning while maintaining the flexibility to generalize across broader domains. Fine-tuning leverages low-rank adaptation (LoRA) techniques, which reduce the number of

trainable parameters, allowing efficient domain adaptation without retraining the entire model [61]. By combining the generalization ability from pre-training with the task-specific precision from fine-tuning, KELMs achieve high performance in both accuracy and knowledge consistency. This dual-phase process ensures that the model not only understands the syntactic and semantic nuances of text but also incorporates and reasons with structured knowledge effectively, making it highly adaptable to diverse real-world applications.

3.3.3 Challenges and optimization directions

The technical layer faces several core challenges in knowledge integration. The semantic gap in retrieval systems creates difficulties where traditional methods like TF-IDF struggle with complex semantic associations, while neural approaches depend too heavily on high-quality annotated data. This can be addressed through hybrid retrieval strategies that combine sparse and dense vectors, along with contrastive learning to optimize embedding spaces and self-supervised tasks such as masked relationship prediction to reduce annotation dependencies.

Knowledge bias in attention mechanisms presents another challenge, creating imbalances in how models utilize available knowledge. By implementing knowledge-aware attention techniques such as entity relationship gating and hierarchical attention mechanisms, models can achieve both global and local focus, effectively directing attention to key logical chains. Additionally, the trade-off between comprehensive fine-tuning and domain transfer poses significant difficulties, as updating all parameters is computationally expensive and risks catastrophic forgetting.

Parameter-efficient fine-tuning approaches, such as LoRA and Adapter, combined with knowledge consistency loss functions, offer promising solutions by enabling domain knowledge injection while preserving general capabilities, as demonstrated in legal applications where only the knowledge adapter layer requires fine-tuning. In the future, the focus will shift to lightweight knowledge injection paradigms and unified multimodal retrieval frameworks that seamlessly integrate text, images, and structured data, reflecting the essential collaboration between knowledge, computation, and task-specific requirements.

3.4 Application layer in KELMs

From the perspective of the application layer, KELMs are designed to solve domain-specific challenges and provide accurate, contextually aware outputs in various fields by leveraging external knowledge, such as knowledge graphs (KGs) or curated databases [12]. These models excel in scenarios requiring high precision, factual grounding, and reasoning capabilities. Below are key application areas and the technologies that support them:

3.4.1 Question answering

KELMs significantly improve question-answering (QA) systems by integrating structured knowledge to generate accurate, contextually relevant, and verifiable responses [62]. When addressing factual queries, such as “What is the capital of France?”, the model relies on a knowledge graph (e.g., DBpedia or Wikidata) to retrieve the required information. Specifically, the model queries the graph using an entity-relation structure, formulated as $q = \text{Retrieve}(h, r)$, where $h = \text{France}$ represents the entity and $r = \text{capital}$ represents the relationship. This process returns the corresponding tail entity $t = \text{Paris}$, providing the answer. By incorporating such external structured knowledge, KELMs ensure high accuracy and consistency, especially in areas like finance, law, and medicine, where accuracy and reliability are paramount [63]. This method not only improves the factual grounding, but also enables reasoning over complex relationships within knowledge graphs, making it a critical capability for QA systems.

3.4.2 Enhanced justification of KG–LLM synergy

The complementary strengths of KGs and LLMs drive AI advances through domain-specific scenarios that require precision, adaptability, and transparency. In healthcare diagnostics, LLMs analyze unstructured patient narratives (e.g. “fatigue and joint pain”) while KGs ground reasoning in verified medical ontologies (e.g., SNOMED-CT relationships like “rheumatoid_arthritis $\rightarrow$ causes $\rightarrow$ synovitis”), reducing diagnostic hallucinations by 42% in MedQA benchmarks. Financial compliance systems like JPMorgan’s COiN platform demonstrate dynamic synergy: LLMs extract temporal transaction patterns (e.g. “frequent \$9500 transfers”) that update AML KGs in real time, achieving 89% accuracy in fraud detection. Educational applications further highlight this symbiosis: when students misunderstand algebraic concepts (e.g. solving “ $3x + 5 = 20$ ” as “ $x = 5$ ”), LLMs generate stepwise corrections while KGs map errors to pedagogical frameworks (e.g., “sign error prevalence: 85%”), enabling Khan Academy’s AI tutor to improve retention by 31%. These scenarios resolve core AI limitations: KGs mitigate LLM hallucination risks (+38% factual consistency), while LLMs democratize KG access through natural language interfaces (75% reduction in SPARQL dependency) - validating their combined role in achieving reliable, scalable, and explainable intelligence.

3.4.3 Challenges and optimization directions

The application layer faces several key challenges. Question-answer systems struggle with credibility when handling complex reasoning chains, despite RAG improvements. Dialogue systems suffer from context drift during extended conversations. Recommendation systems face cold start problems and lack explainability for new users or niche products. Ethical considerations include privacy risks, algorithmic bias, and vulnerability to attacks [64]. To address these challenges, integrating GNNs for multihop reasoning with user feedback mechanisms, implementing dialogue state tracking with dynamic knowledge graph binding, and leveraging implicit

relationships to improve recommendations are efficient methods. Privacy-preserving techniques such as federated learning, bias audit pipelines, and differential privacy can mitigate ethical risks [65].

Current research focuses on exploring knowledge scheduling mechanisms that adaptively integrate domain-specific resources according to operational contexts, aiming to evolve AI systems from task-specific tools into collaborative frameworks with enhanced problem solving capacities.

4 Case study

We use GLM [66] for the case study. GLM is a representative knowledge-enhanced LLM that serves in many different scenarios. In-depth analysis of GLM is structured around four key dimensions: the data layer, model layer, technology layer, and application layer, offering insights into its role of KELM. To address the performance of GLM on knowledge-intensive tasks, experimental results on benchmark datasets, including MedQA and a subset of medical-related MMLU, provide convincing evidence. Independent studies show that the GLM KGs-enhanced variant is significantly better than its base version, with a 15–20% improvement in MedQA and 14–17% in medical MMLU [67]. These improvements stem from the knowledge injection mechanism detailed in Sect. 3.2, which effectively addresses the hallucination problem while preserving the model's linguistic capabilities. In addition, the evaluations of knowledge-centric queries in AlignBench [68] show that the GLM models have adversarial or better performance in factual accuracy and knowledge retrieval than other large language models, especially in specialized domains.

From the data layer, GLM's data foundation comes from the massive text data on the Internet, including books, Wikipedia entries, news articles, and Web documents [69]. These data have been consumed, such as cleaning, reorganization, and word segmentation, to ensure the quality and diversity of the data. GLM uses an autoregressive fill-in-the-blank pretraining task to learn the semantic structure and pattern of language by predicting randomly occluded text fragments [70].

From the perspective of model layer, GLM adopts a variant of Transformer architecture, using a multilayer self-attention mechanism to capture long-distance dependencies in text, and alleviates the gradient problem in deep network training through residual connection and layer normalization [71]. The core innovation of GLM lies in the design of its pre-training task, which predicts the entire masked text segment instead of predicting each masked word independently, thus cultivating the ability to generate coherent text in the pre-training stage.

At the technical layer, GLM significantly improves the performance and accuracy of the model by combining LLM with RAG. GLM adopts a two-stream model architecture. The generation stream is responsible for generating natural language responses based on the context, while the search stream uses databases, knowledge graphs, or document collections for information retrieval to supplement the LLM generation capabilities. GLM can also perform multimodal knowledge integration, extend the retrieval module to images or structured data, and uniformly process different data modes to improve the quality of answers [72].

Table 3 Enhanced comparative analysis of KELMs

Comparison dimension	GPT-3	BERT	RoBERTa	ERNIE	GLM
<i>Data layer</i>					
Data composition	45TB web text	16GB multilingual corpus	160GB filtered web text	5TB Baidu Chinese corpus	Chinese-English scientific literature Dialog corpus
Pretraining strategy	Autoregressive generation	Masked language modeling	Dynamic masking optimization	Entity-relation masking	Two-stage pretraining: Autoregressive + Autoencoding
Knowledge integration	Implicit context correlation	Attention weight integration	Dynamic context correlation	Knowledge graph entity injection	Retrieval-augmented generation Architecture
Training efficiency	FP32 full-precision training	FP32 standard training	FP16 mixed precision	FP16 standard training	Quantized mixed-precision training with memory optimization
<i>Model layer</i>					
Core architecture	175B parameter Transformer	340 M parameter bidirectional Transformer	355 M parameter dynamic Transformer	100B parameter modal architecture	Parameter-efficient architecture with sparse attention
Attention mechanism	Sparse attention	Full self-attention	Dynamic window attention	Entity-relation attention	Efficient multi-head sparse attention with memory saving
Generation paradigm	Unidirectional autoregressive	Masked prediction	Context-conditioned generation	Knowledge-guided generation	Bidirectional generation. Optimization of multi-turn dialog
<i>Technical layer</i>					
Knowledge encoding	Implicit vector storage	Context-aware encoding	Dynamic context encoding	Structured knowledge encoding	Hierarchical knowledge encoding (Concept-entity-relation)
Fine-tuning method	Few-shot learning	Task-specific fine-tuning	Contrastive learning	Knowledge distillation	Incremental pretraining. Multitask joint fine-tuning

Table 3 (continued)

Comparison dimension	GPT-3	BERT	RoBERTa	ERNIE	GLM
Memory Optimization	Basic caching mechanism	No special design	Dynamic memory allocation	Knowledge cache pool	Memory paging management. Gradient checkpointing
<i>Application layer</i>					
Scientific QA	General domain QA	Short-text matching	Context-aware QA	Structured knowledge QA	Multi-granularity reasoning. Formula derivation support
Technical support	Open-domain dialog generation	Intent classification	Context-enhanced dialog	Multi-turn dialog tracking	Knowledge-augmented dialog. Accuracy improvement
Multilingual support	Primarily English	104 languages	100 languages	Chinese-optimized	Chinese-English bilingual optimization. Value enhancement

Finally, in the application layer, GLM shows high flexibility and wide applicability in multiple downstream tasks. It can handle various NLP tasks, including text classification, sentiment analysis, question-answering systems, text summarization, and machine translation. In the fine-tuning stage, GLM adopts the method of reformulating the task as a fill-in-the-blank task, which is not only consistent with the pre-training task, but also improves the task-specific performance, so that the model can naturally adapt to different downstream tasks.

In summary, GLM's innovative design and versatile capabilities across data, model, technical, and application layers position it as a powerful and adaptable pretraining framework, delivering state-of-the-art performance across a wide range of NLP tasks, and demonstrating significant potential for future advancements in the field. Table 3 provides a comparative analysis of knowledge-enhanced large language models, highlighting the unique characteristics and performance metrics of GLM in relation to other prominent models.

5 Conclusion

This study has comprehensively analyzed the synergistic integration of Large Language Models (LLMs) and Knowledge Graphs (KGs) through a systematic four-layer framework. By examining their complementary strengths across the data, model, technology, and application layers, we have demonstrated how this integration addresses the fundamental limitations of each technology while enhancing their combined capabilities. The analysis reveals that LLMs contribute to contextual understanding, natural language generation, and adaptability to unstructured data, while KGs provide structured, verifiable knowledge representations with explicit reasoning paths. We have reviewed the integration strategies, from knowledge injection and retrieval-augmented generation to neuro-symbolic approaches. Integration shows significant potential to create AI systems that are both more capable and more trustworthy.

The GLM case study demonstrates the practical effectiveness of these integration approaches, with substantial performance improvements in knowledge-intensive tasks (15–20% on MedQA and 14–17% on the medical MMLU benchmarks). The results validate our multilayered framework and demonstrate that combining the reasoning capabilities of KGs with the generative power of LLMs can produce more accurate, interpretable, and factually consistent results.

Looking ahead, promising research directions include lightweight knowledge representation, adaptive knowledge update strategies, and unified cross-modal frameworks. Although challenges remain in interdisciplinary integration, security, and cost-effectiveness, the combined KG–LLM approach represents a significant advancement toward more intelligent, reliable, and transparent AI systems capable of addressing complex real-world problems.

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Declarations

Competing Interests The authors declare no competing interests

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